# Giulio Ermanno Pibiri

*Curriculum Vitae et Studiorum*

## Contact Information

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Personal Web page `https://jermp.github.io`

## Personal Information

Place of birth Bagno a Ripoli (Florence), Italy
Date of birth 13 July 1990

## Education

01/11/2015 – 31/10/2018 **PhD in Computer Science (INF/01)**.
- University of Pisa, Pisa, Italy
- Thesis: *Space- and Time-Efficient Data Structures for Massive Datasets* (Defended on 08/03/2019)
- Grade: Excellent
- Supervisor: Rossano Venturini (`https://rossanoventurini.github.io`)

2012 – 2014 **Master Degree in Computer Science & Networking (class LM18)**.
- University of Pisa and Scuola Superiore Sant'Anna, Pisa, Italy
- Thesis: *Dynamic Elias-Fano Encoding* (Defended on 06/03/2015)
- Grade: 110/110 *summa cum laude*
- Supervisor: Rossano Venturini (`https://rossanoventurini.github.io`)

2009 – 2012 **Bachelor Degree in Computer Engineering (class L08)**.
- University of Florence, Florence, Italy
- Thesis: *Quantum Computation & Grover's Algorithm* (Defended on 09/10/2012)
- Grade: 110/110 *summa cum laude*
- Supervisor: Gabriele Vezzosi (`http://www.dma.unifi.it/~vezzosi`)

2004 – 2009 **High School Diploma**.
- Liceo Scientifico Statale Guido Castelnuovo, Florence, Italy
- Grade: 100/100.

## Research Interests

Keywords Data Structures, Data Compression, Indexing, Efficiency

Short Description

My research activity focuses on the design and implementation of compressed data structures for indexing large quantities of data coming from different fields of computing, such as Bioinformatics, Information Retrieval, and Natural Language Processing. The main objective is to improve the efficiency of complex tasks in these fields by providing efficient (i.e., *fast*) and effective (i.e., *small*) indexes to maintain/query data. In fact, a compressed index uses less storage space than the original data, thus permitting:

- for a fixed memory budget, to handle larger datasets;
- for the same dataset, to maintain its compressed representation in faster memory levels (e.g., RAM instead of disk), hence granting faster access.

I am committed to efficient software production: my software is available on GitHub.

Research Problems

*Minimizer sampling schemes* (WABI 2024, AMB 2025); *Robustness Verification of Tree Ensembles* (CSS 2023, S&P 2025); *Colored $k$-mer Indexing* (WABI 2023, RECOMB 2024, AMB 2024, JCB 2024, WABI 2025); *Reference Indexing* (RECOMB 2023); *Locality-Preserving Minimal Perfect Hashing for $k$-mers* (BIOINF 2023); *Compressed and Weighted Dictionaries for $k$-mers* (BIOINF 2022, WABI 2022, AMB 2023, GBIO 2023); *Time Series Compression* (SPIRE 2021); *Minimal Perfect Hashing* (SIGIR 2021, TKDE 2023, ESA 2024); *Prefix-Sums* (SPE 2020); *Rank/Select Queries* (INFOSYS 2021); *Query Auto-Completion* (SIGIR 2020); *Bitmap Compression* (DCC 2021); *Indexing of Semantic Relations* (TKDE 2020, ICDE 2021); *Indexing and Estimation of Language Models* (SIGIR 2017, TOIS 2019); *Inverted Index Compression* (TOIS 2017, EBDT 2018, WSDM 2019, TKDE 2019, CSUR 2020); *Succinct and Dynamic Ordered Sets of Integers* (CPM 2017).

## Research Positions

06/06/2025 – present

**Associate Professor of Computer Science (INF/01)**.

- Department of Environmental Sciences, Informatics and Statistics, "Ca' Foscari" University of Venice, Venice, Italy
- Protocollo n. 0134983, 03/06/2025

06/06/2022 – 05/06/2025

**Assistant Professor of Computer Science (INF/01)**.

- Department of Environmental Sciences, Informatics and Statistics, "Ca' Foscari" University of Venice, Venice, Italy
- Protocollo n. 52696, 01/06/2022
- Scientific Habilitation for Associate Professorship for Computer Science (01/B1) and Computer Engineering (09/H1) earned on 19-th November 2024.

15/03/2021 – 31/05/2022

**Postdoctoral Research Fellow in Computer Science**.

- Institute of Science and Information Technologies "A. Faedo" (ISTI), National Research Council of Italy (CNR), Pisa, Italy
- Research grant issued on the European project ACCORDION with theme "Tecniche algoritmiche per compressione, indicizzazione e ricerca di grandi quantità di dati e progettazione di relative librerie software open source" (Protocollo n. 0000901/2021, 09/03/2021, ISTI 004/2021 - PI)

01/11/2018 – 28/02/2021

**Postdoctoral Research Fellow in Computer Science**.

- Institute of Science and Information Technologies "A. Faedo" (ISTI), National Research Council of Italy (CNR), Pisa, Italy
- Research grant issued on the European project BIGDATAGRAPES with theme "Compressione, indicizzazione e ricerca su grandi collezioni di dati semantici" (Protocollo n. 0003847, 24/10/2018, ISTI 014/2018 - PI)

01/11/2015 – 31/10/2018 **PhD Student in Computer Science**.

- University of Pisa, Pisa, Italy
- Thesis: *Space- and Time-Efficient Data Structures for Massive Datasets*
- Supervisor: Rossano Venturini (`https://rossanoventurini.github.io`)

Part of the research was conducted **abroad** (6 months):

- 01/05/2018 – 01/10/2018
  - The University of Melbourne, School of Computing and Information Systems, Melbourne, Australia
  - Supervisor: Alistair Moffat (`https://people.eng.unimelb.edu.au/ammoffat`)
  - Worked on fast dictionary-based decoding of compressed inverted index data.
- 01/04/2018 – 30/04/2018
  - RIKEN Advanced Intelligence Project (AIP), Tokyo, Japan
  - Supervisor: Yasuo Tabei (`https://sites.google.com/site/yasuotabei`)
  - Worked on various problems, such as, string similarity search, trie indexing, rank/select indexes, and sparse matrix multiplication.

## Research Visits

19/10/2022 – 21/10/2022 Inria, Rennes, France. Visiting dr. Pierre Peterlongo and dr. Karel Brinda.

10/05/2022 – 12/05/2022 The University of Lille, CNRS, CRIStAL Laboratory, Lille, France. Visiting dr. Antoine Limasset and dr. Camille Marchet.

01/05/2018 – 01/10/2018 The University of Melbourne, School of Computing and Information Systems, Melbourne, Australia. Visiting prof. Alistair Moffat and dr. Matthias Petri.

01/04/2018 – 30/04/2018 RIKEN Advanced Intelligence Project (AIP), Tokyo, Japan. Visiting dr. Yasuo Tabei and dr. Shunsuke Kanda.

## Projects

01/09/2021 – 31/05/2021 **Italian PON project OK-INSAID**.

- Institute of Science and Information Technologies "A. Faedo" (ISTI), National Research Council of Italy (CNR), Pisa, Italy
- Role: Investigator

15/03/2021 – 31/05/2021 **European project ACCORDION**.

- Institute of Science and Information Technologies "A. Faedo" (ISTI), National Research Council of Italy (CNR), Pisa, Italy
- Role: Investigator

01/11/2018 – 28/02/2021 **European project BIGDATAGRAPES**.

- Institute of Science and Information Technologies "A. Faedo" (ISTI), National Research Council of Italy (CNR), Pisa, Italy
- Role: *Task Leader* for the task 3.3 "Big Data Indexing"

01/06/2017 – 31/10/2018 **European project LIGA**.

- Institute of Science and Information Technologies "A. Faedo" (ISTI), National Research Council of Italy (CNR), Pisa, Italy
- Role: Principal Developer

## Teaching Activity

A.Y. 2024/2025 Teacher for COMPUTER SCIENCE I - MOD. 1 (code CT0569, 6 CFU), Bachelor Degree in Ingegneria Fisica, Ca' Foscari University of Venice

A.Y. 2024/2025 Teacher for INTRODUCTION TO COMPUTER PROGRAMMING (code CT0665, 6 CFU), Bachelor Degree in Computer Science, Ca' Foscari University of Venice

A.Y. 2024/2025 Teacher for PROGRAMMING AND LABORATORY-2 (code CT0442, 6 CFU), Bachelor Degree in Computer Science, Ca' Foscari University of Venice

A.Y. 2023/2024 Teacher for AUTONOMOUS, DISTRIBUTED AND PERVASIVE SYSTEMS-2 (code PHD156-2, 2 CFU), PhD Program in Computer Science, Ca' Foscari University, Italy

A.Y. 2023/2024 Teacher for COMPUTER SCIENCE I - MOD. 1 (code CT0569, 6 CFU), Bachelor Degree in Ingegneria Fisica, Ca' Foscari University, Italy

A.Y. 2023/2024 Teacher for INTRODUCTION TO PROGRAMMING-1 (code CT0441, 6 CFU), Bachelor Degree in INFORMATICA, Ca' Foscari University, Italy

A.Y. 2023/2024 Teacher for PROGRAMMING AND LABORATORY-2 (code CT0442, 6 CFU), Bachelor Degree in INFORMATICA, Ca' Foscari University, Italy

A.Y. 2022/2023 Teacher for Introduction to Algorithms (3h), Invited lecture, IIS A. Pacinotti, Mestre (Venice), Italy

A.Y. 2022/2023 Teacher for KNOWLEDGE, INTERACTION AND INTELLIGENT SYSTEMS-2 (code PHD157-2, 2 CFU), PhD Program in Computer Science, Ca' Foscari University, Italy

A.Y. 2022/2023 Teacher for COMPUTER SCIENCE I - MOD. 1 (code CT0569, 6 CFU), Bachelor Degree in Ingegneria Fisica, Ca' Foscari University, Italy

A.Y. 2022/2023 Teacher for COMPUTER SCIENCE FOR CULTURAL HERITAGE (code CT0612, 6 CFU), Bachelor Degree in SCIENZE E TECNOLOGIE PER I BENI CULTURALI, Ca' Foscari University, Italy

A.Y. 2022/2023 Teacher for PROGRAMMING AND LABORATORY-2 (code CT0442, 6 CFU), Bachelor Degree in INFORMATICA, Ca' Foscari University, Italy

A.Y. 2021/2022 Teacher for *Theory and Practice of Data Compression* (5 CFU), PhD Program in Ingegneria dell'Informazione, University of Pisa, Italy

A.Y. 2019/2020 Teacher for *Algorithmics and Laboratory - Corso B* (code 008AA, 3 CFU), Bachelor Degree in Computer Science, University of Pisa, Italy

A.Y. 2018/2019 Assistant for *Algorithmics and Laboratory - Corso A* (code 008AA, 3 CFU), Bachelor Degree in Computer Science, University of Pisa, Italy

A.Y. 2017/2018 Assistant for *Competitive Programming and Contests* (code 645AA, 6 hours), Master Degree in Computer Science, University of Pisa, Italy

A.Y. 2016/2017 Assistant for *Competitive Programming and Contests* (code 645AA, 6 hours), Master Degree in Computer Science, University of Pisa, Italy

A.Y. 2015/2016 Teacher for *Algorithmics and Laboratory - Corso di recupero* (code 008AA, 3 CFU), Bachelor Degree in Computer Science, University of Pisa, Italy

A.Y. 2015/2016 Assistant for *Algorithmics and Laboratory - Corso A* (code 008AA, 3 CFU), Bachelor Degree in Computer Science, University of Pisa, Italy

## Awards and Grants

2025 *Young Researcher Award Advanced* for the year 2024, issued by ISTI-CNR.

2025 Distinguished Paper Award for the paper *Verifiable Boosted Tree Ensembles*, published in the 46th IEEE Symposium on Security and Privacy (S&$P$).

2022 *Young Researcher Award* for the year 2021, issued by ISTI-CNR.

2021 *Young Researcher Award* for the year 2020, issued by ISTI-CNR.

2020 *Young Researcher Award* for the year 2019, issued by ISTI-CNR.

2017 *SIGIR Student Travel Grant*, issued by ACM SIGIR.

2015 *PhD Scholarship*, issued by the University of Pisa, Department of Computer Science.

2015 *Master Degree Award* for the A.Y. 2013/2014, issued by Scuola Superiore Sant'Anna.

2015 *Best Master Thesis Award in Theoretical Computer Science*, issued by the Italian chapter of the European Association for Theoretical Computer Science (EATCS).

## Organizing Committees

2025 The 23-rd Symposium on Experimental and Algorithms (SEA).

2023 Co-Chair of the 18-th edition of the Workshop on Compression, Text, and Algorithms (WCTA), co-located with SPIRE 2023.

2020 The 28-th edition of the Annual European Symposium on Algorithms (ESA 2020).

2019 The 30-th edition of the International Symposium on Combinatorial Pattern Matching (CPM 2019).

2017 The 24-th International Symposium on String Processing and Information Retrieval (SPIRE 2017).

2016 The 39-th ACM International SIGIR Conference on Research and Development in Information Retrieval (SIGIR 2016). (As student volunteer.)

## Program Committees

2026 The 34-th International Conference on Intelligent Systems for Molecular Biology (ISMB 2026).

2025 The 32nd International Symposium on String Processing and Information Retrieval (SPIRE 2025).

2025 The 23-rd Symposium on Experimental and Algorithms (SEA 2025).

2025 The 15-th RECOMB Satellite Conference on Biological Sequence Analysis (RECOMB-SEQ 2025).

2024 The 47-th ACM International SIGIR Conference on Research and Development in Information Retrieval (SIGIR 2024).

2024 The 14-th RECOMB Satellite Conference on Biological Sequence Analysis (RECOMB-SEQ 2024).

2023 The 31-st European Symposium on Algorithms (ESA 2023 – Track B).

2023 The 13-rd RECOMB Satellite Conference on Biological Sequence Analysis (RECOMB-SEQ 2023).

2023 The 46-th ACM International SIGIR Conference on Research and Development in Information Retrieval (SIGIR 2023).

2023 The 13-rd International Symposium on Algorithms and Complexity (CIAC 2023).

2023 The 45-th European Conference on Information Retrieval (ECIR 2023).

2023 The 16-th International ACM Conference on Web Search and Data Mining (WSDM 2023).

2022 The 45-th ACM International SIGIR Conference on Research and Development in Information Retrieval (SIGIR 2022).

2022 The 44-th European Conference on Information Retrieval (ECIR 2022).

2022 The 15-th International ACM Conference on Web Search and Data Mining (WSDM 2022).

2021 The 30-th ACM International Conference on Information and Knowledge Management (CIKM 2021).

2021 The 44-th ACM International SIGIR Conference on Research and Development in Information Retrieval (SIGIR 2021).

2021 The 43-rd European Conference on Information Retrieval (ECIR 2021).

2021 The 14-th International ACM Conference on Web Search and Data Mining (WSDM 2021).

2020 The 29-th ACM International Conference on Information and Knowledge Management (CIKM 2020).

2020 The 43-rd ACM International SIGIR Conference on Research and Development in Information Retrieval (SIGIR 2020).

2019 The 42-nd ACM International SIGIR Conference on Research and Development in Information Retrieval (SIGIR 2019).

## Reviewing Activity

2016 – present I am or have been an anonymous reviewer for the following conferences/journals.

### Conferences

- SIGIR – ACM Conference on Research and Development in Information Retrieval
- WSDM – ACM Conference on Web Search and Data Mining
- WWW – The Web Conference
- CIKM – ACM Conference on Information and Knowledge Management
- CPM – Annual Symposium on Combinatorial Pattern Matching
- DCC – IEEE Data Compression Conference
- ECIR – European Conference on Information Retrieval
- ESA – European Symposium on Algorithms
- SPIRE – String Processing and Information Retrieval
- ISAAC – International Symposium on Algorithms and Computation
- RECOMB – International Conference on Research in Computational Molecular Biology
- RECOMB-SEQ – Satellite Conference on Biological Sequence Analysis
- CIAC – International Symposium on Algorithms and Complexity
- SODA – Symposium on Discrete Algorithms

### Journals

- ACM Transactions on Algorithms
- Bioinformatics
- Genome Biology
- Information Systems
- Software: Practice and Experience
- Journal of Experimental Algorithmics
- Algorithmica
- MDPI Algorithms
- IEEE Transactions on Information Forensics and Security
- Information Processing and Management
- PCI Math & Comp. Biol.

## Editorial Service

2024 – present I am serving as a guest editor for the Special Issue "Computation over compressed data" (selected papers from DCC 2024) in Information Systems.

2024 – present I am member of the editorial board of *Information Systems*.

2024 – present I serve as a recommender (similar to an editor) of PCI *Mathematical and Computational Biology*.

## Talks

### Invited

04/07/2025 *Repetition-aware compression and query of colored de Bruijn graphs*. RIKEN-AIP Workshop: Compressed Data Structures for Advanced Data Analysis 2025. Tokyo, Japan.

02/07/2025 *The anatomy of an order-preserving k-mer dictionary*. RIKEN-AIP Workshop: Compressed Data Structures for Advanced Data Analysis 2025. Tokyo, Japan.

22/11/2023 *Compressing and indexing pangenomes with meta-colored compacted de Bruijn graphs*. ALPACA-PANGAIA Annual Workshop. Amsterdam, Netherlands.

20/10/2022 *Modular reference indexing with the de Bruijn graph: overview and challenges*. Workshop on Indexing Omic Sequences. Rennes, France.

10/05/2022 *On Weighted K-Mer Dictionaries*. TUDASTIC 2022 (TUtorials on DAta Structures for Text Indexation and Compression). Lille, France.

### Conferences

22/07/2025 *U-Index: A Universal Indexing Framework for Matching Long Patterns*. SEA 2025. Venice, Italy.

02/09/2024 *The Mod-Minimizer: A Simple and Efficient Sampling Algorithm for Long K-Mers*. WABI 2024. Egham, UK.

30/04/2024 *Meta-Colored Compacted de Bruijn Graphs*. RECOMB 2024, Boston, USA.

25/07/2023 *Locality-Preserving Minimal Perfect Hashing of K-Mers*. ISMB 2023, Lyon, France.

21/03/2023 *Spectrum Preserving Tilings Enable Sparse and Modular Reference Indexing*. DSB 2023, Delft, Netherlands.

05/09/2022 *On Weighted K-Mer Dictionaries*. WABI 2022, Potsdam, Germany.

13/07/2022 *Sparse and Skew Hashing of K-Mers*. ISMB 2022. Madison, Wisconsin, USA.

14/06/2022 *PTHash: Revisiting FCH Minimal Perfect Hashing*. DSB 2022, Düsseldorf, Germany.

20/05/2022 *Sparse and Skew Hashing of K-Mers*. RECOMB-seq 2022. La Jolla, California, USA.

04/10/2021 *TSXor: A Simple Time-Series Compression Algorithm*. SPIRE 2021. Lille, France (Virtual event).

14/09/2021 *PTHash: Revisiting FCH Minimal Perfect Hashing*. IIR 2021. Polytechnic University of Bari, Bari (Virtual event).

07/2021 *PTHash: Revisiting FCH Minimal Perfect Hashing*. SIGIR 2021. Montreal, Canada (Virtual event).

04/2021 *Compressed Indexes for Fast Search of Semantic Data*. ICDE 2021. Chania, Greece (Virtual event).

03/2021 *Fast and Compact Set Intersection through Recursive Universe Partitioning*. DCC 2021. Snow Bird, USA (Virtual event).

27/07/2020 *Efficient and Effective Query Auto-Completion*. SIGIR 2020. Xi'An, China (Virtual event).

17/09/2019 *Compressed Indexes for Fast Search of Semantic Data*. IIR 2019. Department of Information Engineering, Padova, Italy.

12/02/2019 *Fast Dictionary-based Compression for Inverted Indexes*. WSDM 2019. Melbourne Exhibition Center, Melbourne, Australia.

10/08/2017 *Efficient Data Structures for Massive N-Gram Datasets*. SIGIR 2017. Keio Plaza Hotel, Tokyo, Japan.

06/07/2017 *Dynamic Elias-Fano Representation*. CPM 2017. University Library of Warsaw, Warsaw, Poland.

06/06/2017 *Efficient Data Structures for Massive N-Gram Datasets*. IIR 2017. Università della Svizzera Italiana, Lugano, Switzerland.

### Seminars

12/07/2024 *Compressed data structures for big data indexing*. Ca' Foscari University of Venice, Venice, Italy.

02/11/2023 *Compressing and indexing pangenomes with meta-colored compacted de Bruijn graphs*. ISTI-CNR, Pisa, Italy (Virtual event).

19/05/2022 *Sparse and Skew Hashing of K-Mers*. ISTI-CNR, Pisa, Italy (Virtual event).

22/12/2021 *Minimal Perfect Hashing and K-Mer String Dictionaries*. "Ca' Foscari" University of Venice, Venice, Italy (Virtual event).

16/11/2021 *PTHash: Revisiting FCH Minimal Perfect Hashing*. ISTI-CNR, Pisa, Italy (Virtual event).

04/03/2021 *Efficiency for Real-World Applications*. ISTI-CNR, Pisa, Italy (Virtual event).

07/06/2019 *Ordered Set Problems*. ISTI-CNR, Pisa, Italy.

01/02/2019 *Indexing Compressed Data for Fast Retrieval*. University of Pisa, Pisa, Italy.

29/10/2018 *Effective Web Graph Representations*. University of Pisa, Pisa, Italy.

17/05/2018 *On Optimally Partitioning Variable-Byte Index Data*. RMIT University, Melbourne, Australia.

10/04/2018 *Elias-Fano Encoding: a powerful tool for data structure design*. RIKEN AIP, Tokyo, Japan.

21/06/2016 *Elias-Fano Encoding: succinct representation of monotone integer sequences with search operations*. University of Pisa, Pisa, Italy.

## Publications

### Journal Papers

AMB 2025 Ragnar Groot Koerkamp and Daniel Liu and Giulio Ermanno Pibiri. *The open-closed mod-minimizer algorithm*. 2025. Algorithms for Molecular Biology (AMB), pages 17.
DOI: 10.1186/s13015-025-00270-0
ISSN: 1748-7188

JCB 2024 Alessio Campanelli and Giulio Ermanno Pibiri and Jason Fan and Rob Patro. *Where the patterns are: repetition-aware compression for colored de Bruijn graphs*. 2024. Journal of Computational Biology (JCB), pages 29.
DOI: 10.1089/cmb.2024.0714
ISSN: 1557-8666

BIOINF 2024 Moein Karami and Aryan Soltani Mohammadi and Marcel Martin and Barış Ekim and Wei Shen and Lidong Guo and Mengyang Xu and Giulio Ermanno Pibiri and Rob Patro and Kristoffer Sahlin. *Designing efficient randstrobes for sequence similarity analyses*. 2024. Bioinformatics (BIOINF), pages 9.
DOI: 10.1093/bioinformatics/btae187
ISSN: 1367-4811

AMB 2024 Jason Fan, Noor Pratap Singh, Jamshed Khan, Giulio Ermanno Pibiri, Rob Patro. *Fulgor: A fast and compact $k$-mer index for large-scale matching and color queries*. 2024. Algorithms for Molecular Biology (AMB), pages 21.
DOI: 10.1186/s13015-024-00251-9
ISSN: 1748-7188

TKDE 2023 Giulio Ermanno Pibiri and Roberto Trani. *Parallel and External-Memory Construction of Minimal Perfect Hash Functions with PTHash*. 2023. IEEE Transactions on Knowledge and Data Engineering (TKDE), pages 12.
DOI: 10.1109/TKDE.2023.3303341
ISSN: 1041-4347

GBIO 2023 Sebastian Schmidt, Shahbaz Khan, Jarno Alanko, Giulio Ermanno Pibiri, Alexandru I. Tomescu. *Matchtigs: minimum plain text representation of $k$-mer sets*. 2023. Genome Biology (GBIO), pages 33.
DOI: 10.1186/s13059-023-02968-z
ISSN: 1474-7596

BIOINF 2023 Giulio Ermanno Pibiri, Yoshihiro Shibuya, and Antoine Limasset. *Locality-Preserving Minimal Perfect Hashing of $k$-mers*. 2023. Bioinformatics (BIOINF), pages 9.
DOI: 10.1093/bioinformatics/btad219
ISSN: 1367-4803

AMB 2023 Giulio Ermanno Pibiri. *On Weighted $k$-mer Dictionaries*. 2023. Algorithms for Molecular Biology (AMB), pages 25.
DOI: 10.1186/s13015-023-00226-2
ISSN: 1748-7188

BIOINF 2022 Giulio Ermanno Pibiri. *Sparse and Skew Hashing of $k$-mers*. 2022. Bioinformatics (BIOINF), pages 9.
DOI: 10.1093/bioinformatics/btac245
ISSN: 1367-4803

INFOSYS 2021 Giulio Ermanno Pibiri and Shunsuke Kanda. *Rank/Select Queries over Mutable Bitmaps*. 2021. Information Systems (INFOSYS), pages 15.
DOI: 10.1016/j.is.2021.101756
ISSN: 0306-4379

CSUR 2020 Giulio Ermanno Pibiri and Rossano Venturini. *Techniques for Inverted Index Compression*. 2020. ACM Computing Surveys (CSUR), pages 36.
DOI: 10.1145/3415148
ISSN: 0360-0300

SPE 2020 Giulio Ermanno Pibiri and Rossano Venturini. *Practical Trade-Offs for the Prefix-Sum Problem*. 2020. Software: Practice and Experience (SPE), pages 29.
DOI: 10.1002/spe.2918
ISSN: 0038-0644

TKDE 2020 Raffaele Perego, Giulio Ermanno Pibiri and Rossano Venturini. *Compressed Indexes for Fast Search of Semantic Data*. 2020. IEEE Transactions on Knowledge and Data Engineering (TKDE), pages 12.
DOI: 10.1109/TKDE.2020.2966609
ISSN: 1041-4347

TKDE 2019 Giulio Ermanno Pibiri and Rossano Venturini. *On Optimally Partitioning Variable-Byte Codes*. 2019. IEEE Transactions on Knowledge and Data Engineering (TKDE), pages 12.
DOI: 10.1109/TKDE.2019.2911288
ISSN: 1041-4347

TOIS 2019 Giulio Ermanno Pibiri and Rossano Venturini. *Handling Massive N-Gram Datasets Efficiently*. 2019. ACM Transactions on Information Systems (TOIS), pages 41.
DOI: 10.1145/3302913
ISSN: 1046-8188

TOIS 2017 Giulio Ermanno Pibiri and Rossano Venturini. *Clustered Elias-Fano Indexes*. 2017. ACM Transactions on Information Systems (TOIS), volume 2, pages 33.
DOI: 10.1145/3052773
ISSN: 1046-8188

## Conference Papers

WABI 2025 Alessio Campanelli and Giulio Ermanno Pibiri and Rob Patro. *Fast pseudoalignment queries on compressed colored de Bruijn graphs*. 2025. The 25-th International Conference on Algorithms in Bioinformatics (WABI), 22 pages.
DOI: 10.4230/LIPIcs.WABI.2025.6
ISBN: 978-3-95977-386-7

SEA 2025 Lorraine A. K. Ayad and Gabriele Fici and Ragnar Groot Koerkamp and Grigorios Loukides and Rob Patro and Giulio Ermanno Pibiri and Solon P. Pissis. *U-index: A Universal Indexing Framework for Matching Long Patterns*. 2025. The 23rd Symposium on Experimental Algorithm (SEA), 18 pages.
DOI: 10.4230/LIPIcs.SEA.2025.4
ISBN: 978-3-95977-375-1

S&$P$ 2025 Stefano Calzavara, Lorenzo Cazzaro, Claudio Lucchese, and Giulio Ermanno Pibiri. *Verifiable Boosted Tree Ensembles*. 2025. The 46th IEEE Symposium on Security and Privacy (S&$P$), 15 pages.
DOI: 10.1109/SP61157.2025.00022
ISBN: 979-8-33152-236-0

WABI 2024 Ragnar Groot Koerkamp and Giulio Ermanno Pibiri. *The mod-minimizer: a simple and efficient sampling algorithm for long $k$-mers*. 2024. International Workshop on Algorithms in Bioinformatics (WABI), 22 pages.
DOI: 10.4230/LIPIcs.WABI.2024.11
ISBN: 978-3-95977-340-9

ESA 2024 Hermann, Stefan, Hans-Peter Lehmann, Giulio Ermanno Pibiri, Peter Sanders, and Stefan Walzer. *PHOBIC: Perfect Hashing with Optimized Bucket Sizes and Interleaved Coding*. 2024. European Symposium on Algorithms (ESA), 17 pages.
DOI: 10.4230/LIPIcs.ESA.2024.69
ISBN: 978-3-95977-338-6

RECOMB 2024 Giulio Ermanno Pibiri, Jason Fan, and Rob Patro. *Meta-colored compacted de Bruijn graphs*. 2024. International Conference on Research in Computational Molecular Biology (RECOMB), 16 pages.
DOI: 10.1007/978-1-0716-3989-4_9
ISBN: 978-1-07163-989-4

CCS 2023 Stefano Calzavara, Lorenzo Cazzaro, Giulio Ermanno Pibiri, and Nicola Prezza. *Verifiable Learning for Robust Tree Ensembles*. 2023. ACM SIGSAC Conference on Computer and Communications Security (CCS), 16 pages.
DOI: 10.1145/3576915.3623100
ISBN: 978-1-45039-450-5

WABI 2023 Jason Fan, Noor Pratap Singh, Jamshed Khan, Giulio Ermanno Pibiri, and Rob Patro. *Fulgor: A fast and compact $k$-mer index for large-scale matching and color queries*. 2023. International Workshop on Algorithms in Bioinformatics (WABI), 21 pages.
DOI: 10.4230/LIPIcs.WABI.2023.18
ISBN: 978-3-95977-243-3

RECOMB 2023 Jason Fan, Jamshed Khan, Giulio Ermanno Pibiri, and Rob Patro. *Spectrum preserving tilings enable sparse and modular reference indexing*. 2023. International Conference on Research in Computational Molecular Biology (RECOMB), 20 pages.
DOI: 10.1007/978-3-031-29119-7_2
ISBN: 978-3-03129-119-7

WABI 2022 Giulio Ermanno Pibiri. *On Weighted $k$-mer Dictionaries*. 2022. International Workshop on Algorithms in Bioinformatics (WABI), 20 pages.
DOI: 10.4230/LIPIcs.WABI.2022.9
ISBN: 978-3-95977-243-3

SPIRE 2021 Andrea Bruno, Franco Maria Nardini, Giulio Ermanno Pibiri, Roberto Trani, and Rossano Venturini. *TSXor: A Simple Time Series Compression Algorithm*. 2021. International Symposium on String Processing and Information Retrieval (SPIRE), 8 pages.
DOI: 10.1007/978-3-030-86692-1_18
ISBN: 978-3-03086-692-1

SIGIR 2021 Giulio Ermanno Pibiri and Roberto Trani. *PTHash: Revisiting FCH Minimal Perfect Hashing*. 2021. ACM Conference on Research and Development in Information Retrieval (SIGIR), pages 10.
DOI: 10.1145/3404835.3462849
ISBN: 978-1-45038-037-9

DCC 2021 Giulio Ermanno Pibiri. *Fast and Compact Set Intersection through Recursive Universe Partitioning*. 2021. IEEE Data Compression Conference (DCC), pages 10.
DOI: 10.1109/DCC50243.2021.00037
ISBN: 978-1-66540-333-7

SIGIR 2020 Simon Gog, Giulio Ermanno Pibiri and Rossano Venturini. *Efficient and Effective Query Auto-Completion*. 2020. ACM Conference on Research and Development in Information Retrieval (SIGIR), pages 10.
DOI: 10.1145/3397271.3401432
ISBN: 978-1-45038-016-4

WSDM 2019 Giulio Ermanno Pibiri, Matthias Petri, and Alistair Moffat. *Fast Dictionary-based Compression for Inverted Indexes*. 2019. ACM Conference on Web Search and Data Mining (WSDM), pages 9.
DOI: 10.1145/3289600.3290962
ISBN: 978-1-45035-940-5

SIGIR 2017 Giulio Ermanno Pibiri and Rossano Venturini. *Efficient Data Structures for Massive N-Gram Datasets*. 2017. ACM Conference on Research and Development in Information Retrieval (SIGIR), pages 10.
DOI: 10.1145/3077136.3080798
ISBN: 978-1-45035-022-8

CPM 2017 Giulio Ermanno Pibiri and Rossano Venturini. *Dynamic Elias-Fano Representation*. 2017. Annual Symposium on Combinatorial Pattern Matching (CPM), pages 14.
DOI: 10.4230/LIPIcs.CPM.2017.30
ISBN: 978-3-95977-039-2

### Posters

ICDE 2021 Raffaele Perego, Giulio Ermanno Pibiri and Rossano Venturini. *Compressed Indexes for Fast Search of Semantic Data*. 2021. IEEE International Conference on Data Engineering (ICDE), pages 2.
DOI: 10.1109/ICDE51399.2021.00248
ISBN: 978-1-72819-185-0

### PhD Thesis

2019 Giulio Ermanno Pibiri. *Space- and Time-Efficient Data Structures for Massive Datasets*. 2019. Ph.D. Thesis, University of Pisa, 210 pages.

### Chapters

EBDT 2018 Giulio Ermanno Pibiri and Rossano Venturini. *Inverted Index Compression*. 2018. Encyclopedia of Big Data Technologies (EBDT), pages 8.
DOI: 10.1007/978-3-319-63962-8_52-1
ISBN: 978-3-31963-962-8

## Software

GitHub profile All software is open-source and available from `https://github.com/jermp`.

### Data Structures

Efficient C++ implementations of the following data structures (see also related publications):

- Inverted Indexes (TOIS 2017, TKDE 2019, WSDM 2019, SIGIR 2020, CSUR 2020)
- Tries (SIGIR 2017, TOIS 2019, TKDE 2020)
- Compressed Bitmaps (DCC 2021)
- Mutable Bitmaps with Rank/Select (INFOSYS 2021)
- Segment-Trees and Fenwick-Trees (SPE 2020)
- Minimal Perfect Hash Functions (SIGIR 2021, TKDE 2023, ESA 2024), Locality-Preserving Minimal Perfect Hash Functions (BIOINF 2023)
- Dictionaries for $k$-mers (BIOINF 2022, WABI 2022, AMB 2023, GBIO 2023)
- Colored $k$-mer indexes (WABI 2023, AMB 2024, RECOMB 2024, JCB 2024, WABI 2025)

A more detailed list follows below.

**minimizers** A collection of minimizer-based sampling algorithms.
Reference publication: WABI 2024, AMB 2025.

**fulgor** Fulgor: A fast and compact $k$-mer index for large-scale matching and color queries.
Reference publication: RECOMB 2023, WABI 2023, AMB 2024, RECOMB 2024, JCB 2024, WABI 2025.

**sshash** SSHash: A compressed, weighted, associative, exact dictionary for k-mers. (See also the membership-only version **sshash-lite**.)
Reference publication: BIOINF 2022, WABI 2022, AMB 2023, GBIO 2023.

**lphash** LPHash: Fast and compact locality-preserving minimal perfect hashing for $k$-mer sets.
Reference publication: BIOINF 2023.

**pthash** PTHash: Fast and compact minimal perfect hash functions.
Reference publications: SIGIR 2021, TKDE 2023, ESA 2024.

**rank_select** Mutable bitmaps with support for Rank and Select queries.
Reference publication: INFOSYS 2021.

**psds** A range of tree-shaped data structures for maintaining prefix-sums, including:
- binary Segment-Tree (top-down and bottom-up),
- b-ary Segment-Tree,
- Fenwick-Tree,
- b-ary Fenwick-Tree,
- blocked Fenwick-Tree,
- truncated Fenwick-Tree.

Reference publication: SPE 2020.

**autocomplete** Efficienct and effective autocompletion framework, based on forward/inverted indexes, succinct RMQ, and string dictionaries (Front-Coding and tries).
Reference publication: SIGIR 2020.

**2i_bench** A benchmarking suite for inverted index data structures, featuring the following compressors:
- Elias-Fano and partitioned Elias-Fano,
- Opt-PFor-Delta,
- Binary Interpolative,
- QMX,
- Simple family,
- Variable-Byte family, including Opt-VByte,
- Gamma, Delta, Rice, Zeta,
- DINT.

Reference publication: CSUR 2020.

**interp** An efficient implementation of the Binary Interpolative Coding algorithm.

**s_indexes** Compressed bitmap indexes that support fast intersection and union.
Reference publication: DCC 2021.

**rdf_indexes** Trie-based indexes for semantic data like RDF triples.
Reference publication: TKDE 2020.

**dint** DINT: fast and compact dictionary-based decoder for inverted lists.
Reference publication: WSDM 2019.

**opt_vbyte** Optimal partitioning of inverted lists compressed using binary vectors and point-wise encoders, like Variable-Byte.
Reference publication: TKDE 2019.

**tongrams** Fast language model queries and estimation in compressed space.
Reference publications: SIGIR 2017, TOIS 2019.

**clustered_indexes** Clustered Elias-Fano inverted indexes.
Reference publication: TOIS 2017.

## Miscellanea

**essentials** A C++ library providing essential core utilities for data structure design and benchmarking. More precisely:
- benchmarking facilities, including: messages displaying local time, configurable timer class, function to prevent code elision by compiler, simple creation and printing of json documents;
- functions to serialize-to and load-from disk data structures,
- functions to compute the number of bytes consumed by data structures,
- support for creating, removing, and iterate inside directories,
- transparent support for contiguous memory allocation.

**cmd_line** Command line parser for C++17. It offers all handy features in just 150 lines of code.

**mm_file** A self-contained, header-only, implementation of memory-mapped files in C++ for both reading and writing.

## Languages

| | | |
|---|---|---|
| Italian | **Native** | *CEFR level: C2* |
| English | **Fluent** | *CEFR level: C1* |
| 2018 | **TOEFL iBT in English**.<br>100 (HIGH level) | |
| 2008 | **First Certificate in English (Level B2)**.<br>University of Cambridge, Cambridge, United Kingdom | |